Solution to analysis in Home Assignment 2

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Analysis

In this report I will present my independent analysis of the questions related to home assignment 2. I have discussed the solution with Mubina Crnic, but I swear that the analysis written here are my own.

Scenario 1 – A first Kalman filter and its properties

In this question the properties of the Kalman filter in a simple one-dimensional case (all variables are scalars) will be considered having the following linear and Gaussian state space model:

$$x_k = x_{k-1} + q_{k-1} \tag{1}$$

$$y_k = x_k + r_k \tag{2}$$

where the motion and measurement noises, q_{k-1} and r_k, are zero-mean Gaussian random variables with variances $Q = 1.5$ and $R = 3$, respectively, and the initial prior is $p(x_0) = \mathcal{N}(x_0; 2, 8)$.

a)

We can see there is kind of pattern where the general motion is the same, which means x and y are the same with exception that y has got some noise which will cause y to differ a bit. However, the difference is still within the 3 standard deviation interval $x \pm 3\sigma$.

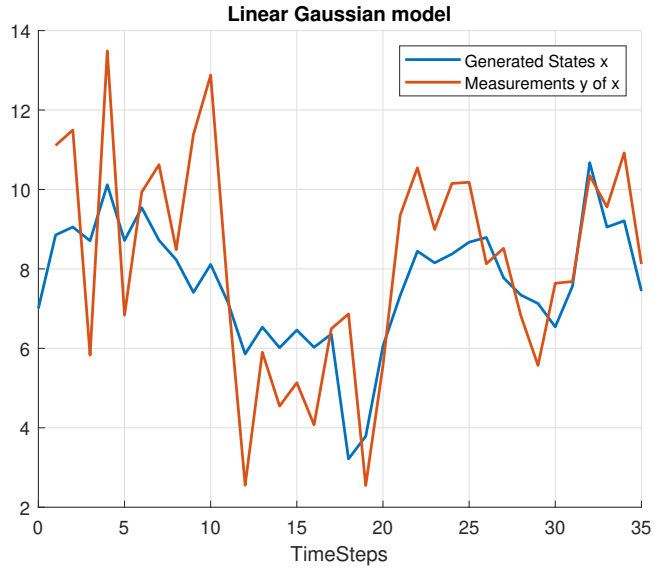


Figure 1: Plot of state sequence and measurement sequence for $N = 35$

b)

We can see that the filter estimation are acceptable since they are very close to the real states x and measurements y. We can also observe that the error covariance matrix is also a great fit since it accounts for uncertainty in the Kalman estimation. This can be observed from that the real states and measurements lie $\hat{x_k} \pm 3\sigma$ meaning that it is inside the 99.7% confidence interval.

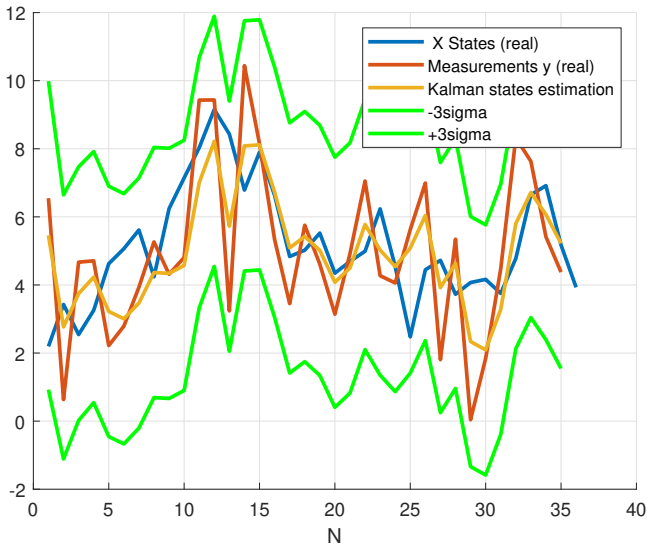


Figure 2: Plot of state sequence, measurement sequence, estimated measurement sequence and sequence of estimates together with the $\hat{x}_k \pm 3\sigma$ level.

When plotting the error density for the time instances $[1, 2, 4, 30]$ around the zero mean, it becomes apparent that the error covariance remains large for lower values of k (fewer time steps), indicating insufficient time for convergence. This observation is reflected in the bandwidth of the respective graphs, where $k = 1$ exhibits a larger bandwidth compared to $k = 30$. A larger bandwidth implies a shorter peak for the probability density function in order to maintain an integral of 1. The physical explanation for it is that as time goes on the Kalman filter will get more inputs to further improve the estimation which leads to a decrease in the uncertainty (sigma gets reduced, more shrunken bandwidth). This process iteratively keeps on happening until convergence of P and K has been achieved.

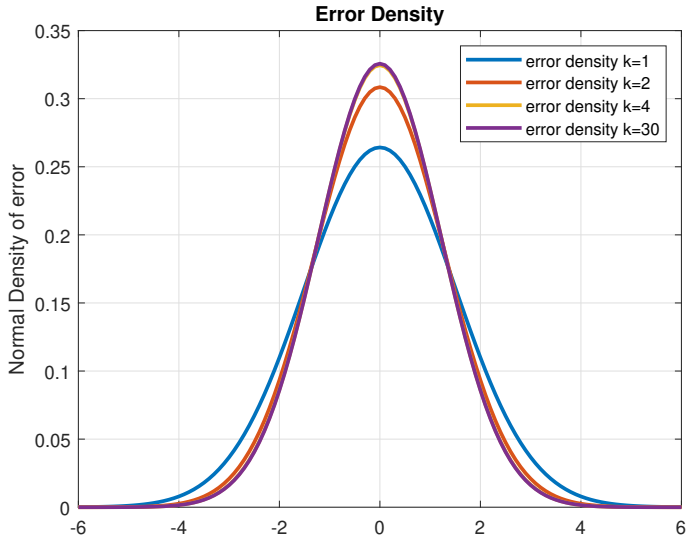


Figure 3: Plot of the error density around zero-mean for time instances $k = [1, 2, 4, 30]$

c)

It this question we can observe the effect of initial prior on the estimation of kalman. We can see when Kalman is given the correct prior it is closer to the true states in the beginning while the kalman with the wrong prior (mean 8) is worse in the beginning. However, it is noteworthy that with time both kalman estimations overlap because we get more measurements into the filter which improves the estimation until convergence.

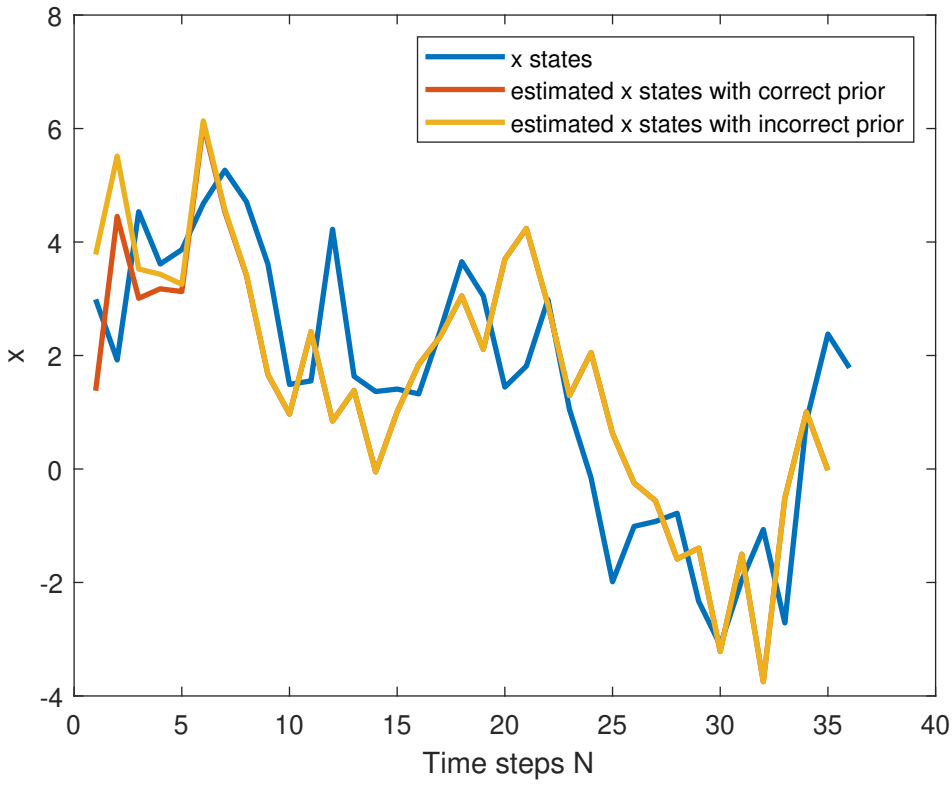


Figure 4: Plot of state sequence, sequence of estimates $\hat{x}_{k|k}$ for correct prior, sequence of estimates $\hat{x}_{k|k}$ for incorrect prior

d)

To address this question, we can gain an approximate understanding of how the outcome might appear in a three-dimensional (3D) context. Initially, we engage in density prediction: by projecting forward without incorporating new data, our uncertainty increases, leading to larger uncertainties. The prediction is done by using the following equations

$$\hat{x}_{k|k-1} = A \cdot \hat{x}_{k-1|k-1}$$
$$P_{k|k-1} = A \cdot P_{k-1|k-1} \cdot A^T + Q$$

Subsequently, we employ measurement updates, wherein we consider our predicted density as the prior and our measurement model as the likelihood (or mea-

surement y). We then combine our predicted density with our likelihood (or new measurement yk) using a factor proportional to our updated posterior density (in kalman filter, this factor is called the Kalman gain K_k to determine $P(x_k|k)$. In this scenario, as our measurement provides informative data, the updated density aligns closely with the observation and diverges from the predicted density. As we receive additional information, uncertainty diminishes.

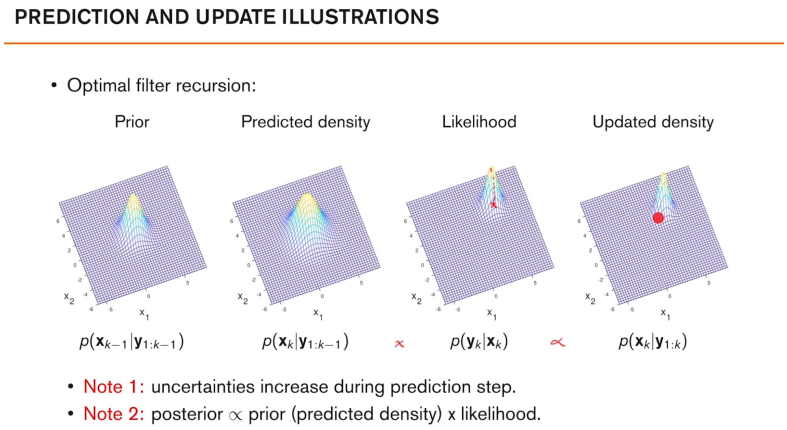


Figure 5: $p(x_{k-1}|y_{1:k-1})$, $p(x_k|y_{1:k-1})$, y_k, and $p(x_k|y_{1:k})$. from lecture slides

Now for our specific case we can observe from the following plot that our result coincides with our conclusion for kalman gain K = 20. What we can see is that

The prediction has larger uncertainty (variance) than the prior however with the same mean. When we measure a new measurement yk it will have a large variance because we have uncertainty from the sensor. Finally for the update step $P(x_k|k)$ will have less uncertainty (σ) than both the prediction and the measurement. It will matter of fact have a very close (if not same) σ as our prior $P(x_{k-1}|k-1)$ since both estimate x at a step with the same amount of respective measurement (either k or k-1). However the mean of the prior and the update is not the same and it is expected since we got a new measurement which has a negative mean which will lead to that the update will move more to the left.

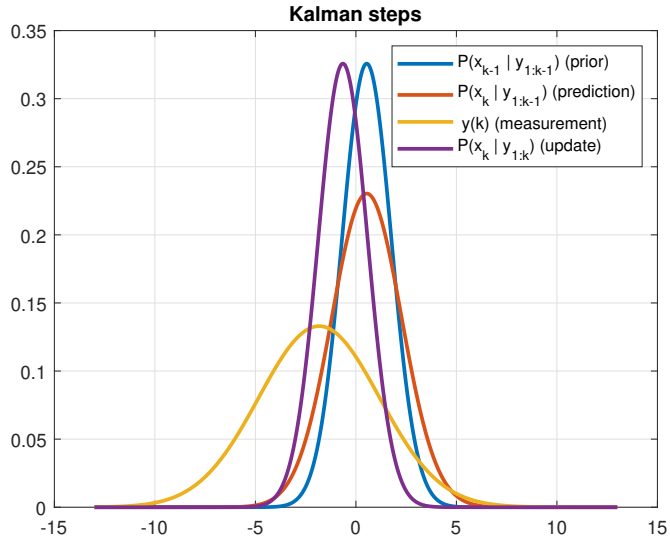


Figure 6: $p(x_{k-1}|y_{1:k-1})$, $p(x_k|y_{1:k-1})$, y_k, *and* $p(x_k|y_{1:k})$ *when* $k = 20$.

e)

In this question we should test the consistency of the kalman filter in two different ways. The long true state sequence of length 110000.

The first way to judge the consistency of the kalman filter: in the following plot we can see a histogram of estimation error $x_k - \hat{x}_{k|k}$ which is almost entirely inside the analytical norm plot $N(0, P_{N|N})$. From this plot we can draw conclusion that the kalman filter is providing is estimating x quite well since the most frequent value of the estimation error (mean) is 0 and the estimation error is quite close to 0 (meaning x = xestimation) in general since 99.97% of the estimation error values lie within 3 standard deviations, meaning the estimation error follows a normal Gaussian distribution.

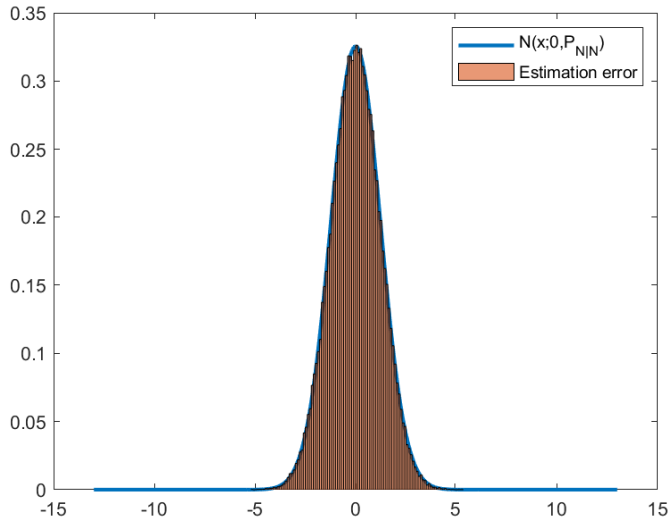


Figure 7: Plot of histogram of the estimation error $x_k - \hat{x}_{k|k}$ *together with PDF* $\mathcal{N}(x; 0, P_{N|N})$ *for* $N = 110000$

There are two other ways which are actually better ways of judging the performance of the kalman filter because there is no need to know what the real value of Xk is (which is the one that we are actually trying to estimate and is hard to know irl). this is why this way is better than the previous way since it needed to know what xk was.

First one is to do some auto correlation of the innovation

$$v_k = y_k - H_k \cdot \hat{x}_{k|k-1}$$

We have at lag 0 the value almost 1 while the other lags we have very small values which means that the innovation process is quite uncorrelated over time which is a great indicator from the filter if we were to auto correlation values which are not significant for the other lags then we would have said the innovation is highly correlated which means the filter has errors bugs.

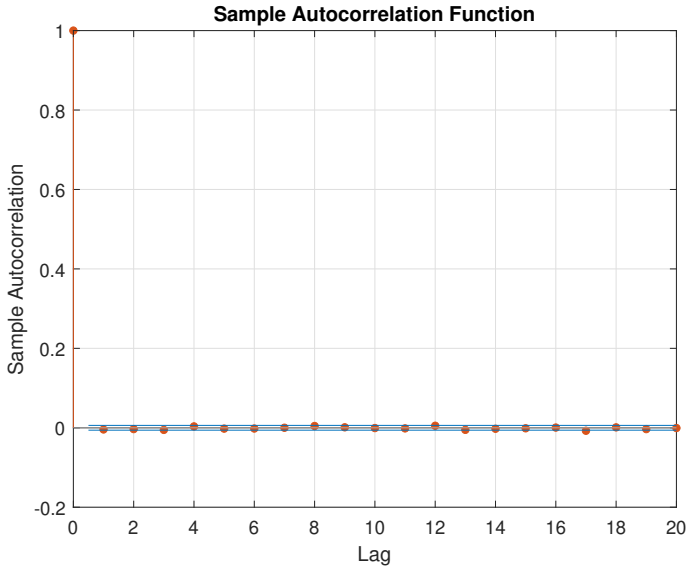


Figure 8: Auto correlation function of the innovation process, v_k

The third way is the so called 3 sigma test (according to me hehe), from the lecture notes we see the following

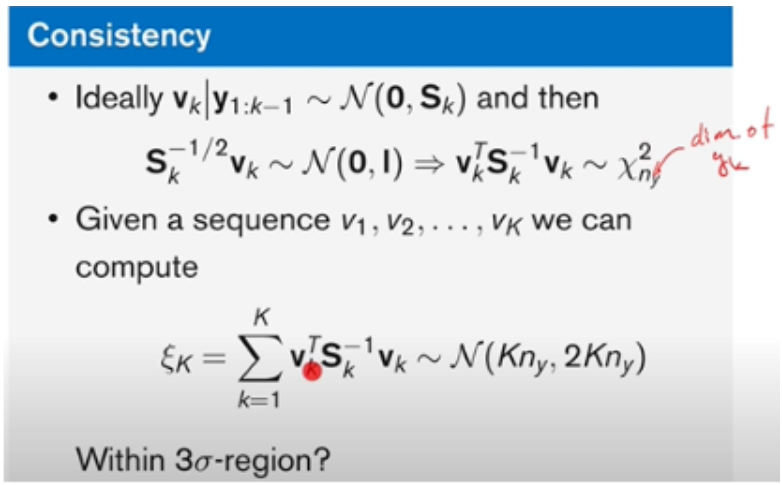


Figure 9: $\pm 3\sigma$ consistency test

To check that this holds true is by constructing a new random variable where we scale the innovation by dividing it by the square root of the covariance. By

doing so we construct a new random variable which is a Gaussian variable with zero mean but a covariance that is an identity matrix. What we do is that we normalize the innovation and transform it into a zero mean Gaussian with an identity matrix as covariance which is easier to interpret. We can use it to design a test statistic which is a wtf chi-squared random variable with the same degree of freedom as the mention of the measurement.

We can then test if the sum of these are distributed according to the distribution written after the sum. Checking if epsilon is within the 3sigma region of the normal distribution written after the sum. Test it visualize by plotting the normalized innovation sequence and see if its zero mean but MORE importantly that the most values lie within +- 3sigma.

From the following plot we can see that the Kalman filter almost values are inside the +3 -3 sigma.

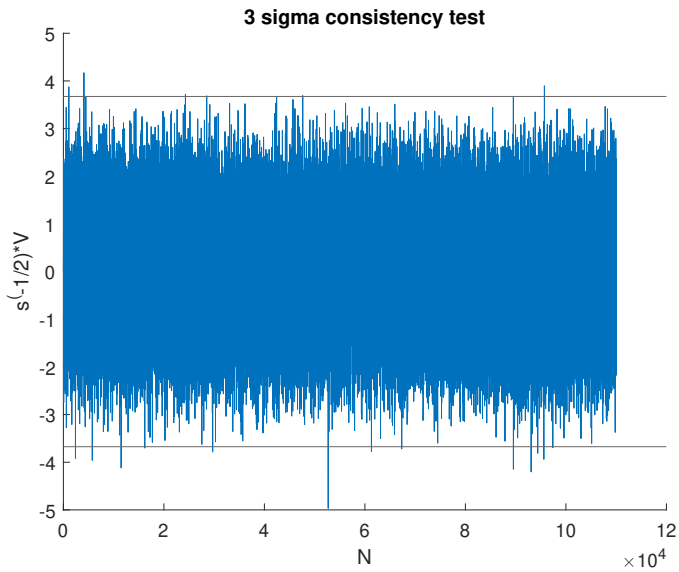


Figure 10: Plot of the $\pm 3\sigma$ consistency test

Scenario 2 – Tuning a Kalman filter

a) - Calibrate speed sensor model

$$y_k = C(v_k + r_k^v)$$

Firstly to determine the value of for the different cases we gotta manipulate the previous equation and by using the assumption that the expected value (mean) of the noise R is 0. Which means C is determined by the following

$$\frac{E[y_k]}{Cv_k}$$

$$C_{stationary} = 1$$

$$C_{v10} = 1.1026$$

$$C_{v20} = 1.1023$$

Now to determine the variance of r

$$Var[r_k^v]$$

we first have to determine the variance of y

$$\sigma_y^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2$$

If you have a sample from a larger population and you want to estimate the population variance, you would divide by $\frac{1}{N-1}$ instead of $\frac{1}{N}$. This is known as Bessel's correction.

Now to determine the covariance of

$$Var[r_k^v]$$

we know that since the first part part of

$$y_k = C(v_k + r_k^v)$$

is a constant (no variation), the variance of σ_y^2 will only depend on the variance of noise R multiplied by C. Now using a proved relation from the previous assignment we will know that

$$\text{Cov}[y] = C \cdot \text{Cov}[R] \cdot C^T$$

and C is a scalar so we have

$$Var[r_k^v] = \frac{\text{Cov}[Y]}{C^2}$$

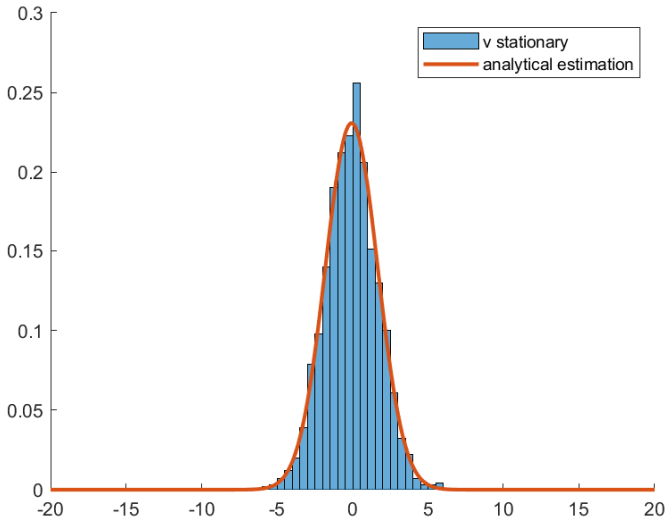


Figure 11: stationary V

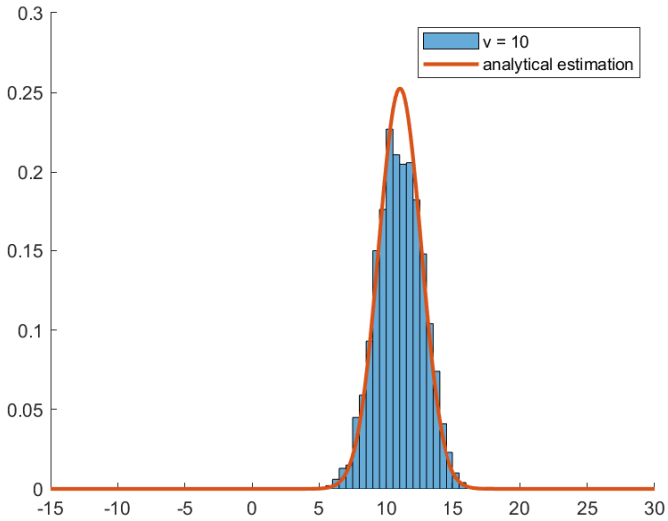


Figure 12: V = 10

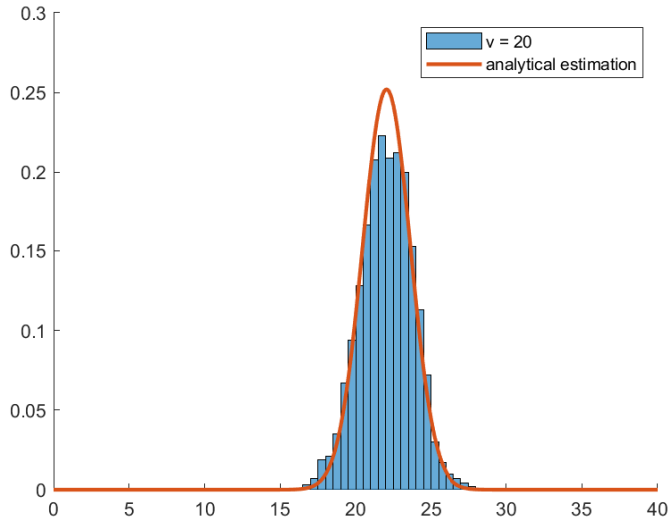


Figure 13: $V = 20$

Finally from the above graphs we can see that our estimation of the sensor parameters which helped us create an analytical estimation which is a normal Gaussian distribution $N \sim (E[y], Var[r_k^v])$.

we can see that it fits respective data information since most of the data lie under our curves for respective case.

b) - Fusing sensors with different rates

The solution to this problem is basically to discard the inputs (columns of data) which has NAN values for position. Doing this makes the sampling intervall of the velocity sensor twice as slow from 0.1s to 0.2s. This will fuse the sensor of the train quite well.

```
% this part is for question 2b
Y_valid=[];
for i=1:length(Y)
   if ~isnan(Y(:,i))
     Y_valid=[Y_valid,Y(:,i)];
   end
end

Y = Y_valid;
```

Figure 14: V = 20

c

In this section we should test two motion model, constant velocity and constant acceleration
Initial xpoint for constant velocity model:

$$x0_{cv} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Initial covariance for constant velocity model:

$$P0_{cv} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Initial xpoint for constant acceleration model:

$$x0_{ca} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

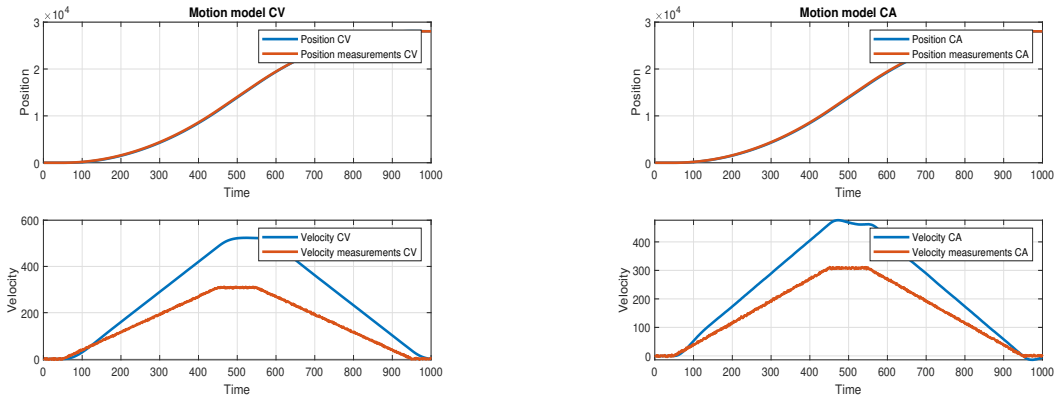
Initial covariance for constant acceleration model:

$$P0_{ca} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The position variance of measurement noise is 2 because the variance of the measurement noise for the velocity was between 2-3. So it was thought that they be set to the same value.

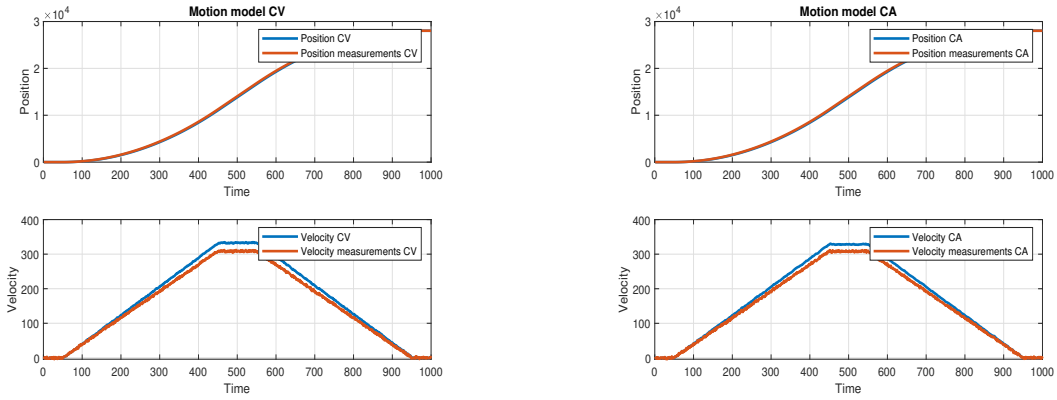
So the only tuning parameters for respective model is $Qv_{continous}$ and $Qa_{continous}$ (the variance of the noise of respective motion model) while the other parameters were set to be fixed

These were tuned or (simulated for different values [0.001, 1, .1, 1000].



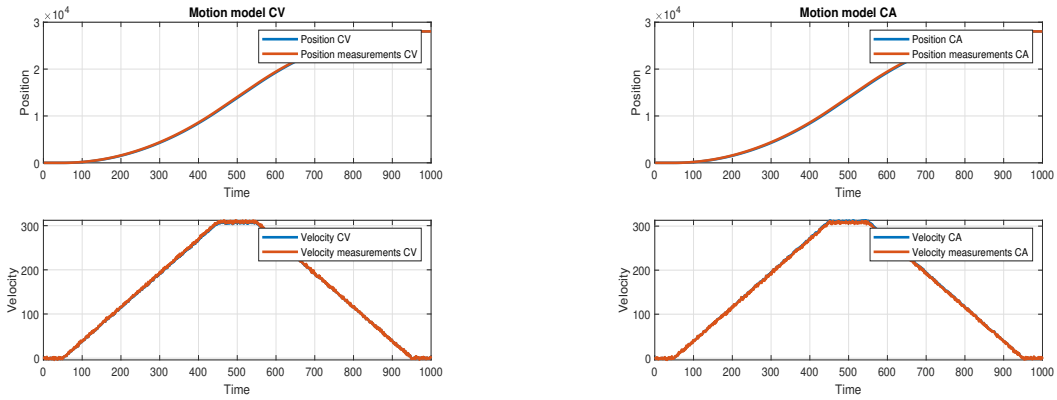
(a) CV, Q = 0.001 *(b) CA, Q = 0.001*

Figure 15: CV and CA for Q = 0.0001



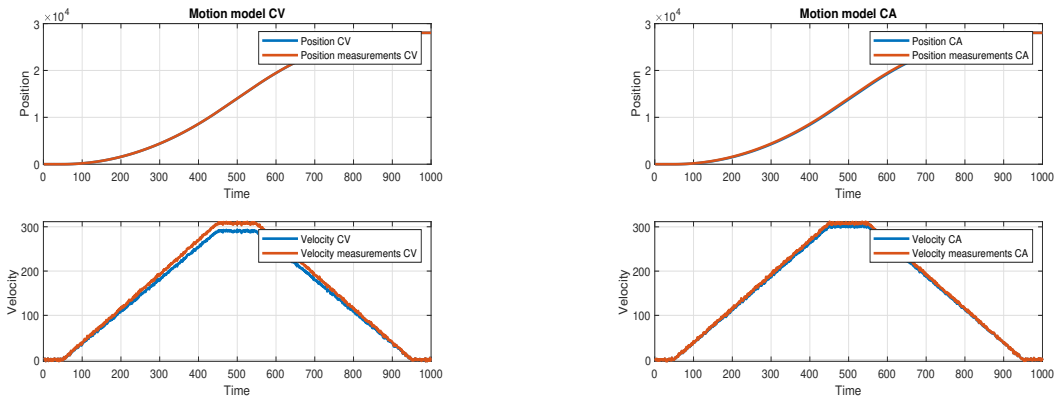
(a) CV, Q = 1 *(b) CA, Q = 1*

Figure 16: CV and CA for Q = 1



(a) CV, Q = 10 *(b) CA, Q = 10*

Figure 17: CV and CA for Q = 10



(a) CV, Q = 1000 *(b) CA, Q = 1000*

Figure 18: CV and CA for Q = 1000

If the the motion model noise variance is small (first case and partially second case), then it means that kalman filter should trust the motion model more the than our measurements and as it can be seen from the graphs. This is bad because if our model is not correct our estimations will be pretty bad, this is specially the case for CV model while CA model is a bit better.

If the the motion model noise variance is close to the noise of the measurement noise, here we are hitting the jackpot between trusting both the measurement models and the motion model which explains why our estimations fit the real data

the best for both CV and CA models.

If the the motion model noise variance is too large it means we should trust the measurement model more which isn't ideal and this can be seen from the graphs were it became worse specially for velocity estimation for CV model while CA model kept it great.